Parity of an infinitely nested series reversion

A proof of a conjecture in OEIS A195194

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Abstract

The sequence A195194 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Paul D. Hanna, is defined by an infinitely nested expression,

$$A(x) = \mathrm{INV}\Big(x - x\,\mathrm{INV}\big(x - 2x\,\mathrm{INV}(x - 3x\,\mathrm{INV}(x - \cdots))\big)\Big),$$

where INV denotes series reversion, the operation that undoes a substitution of power series. Its terms are $1, 1, 4, 29, 312, 4528, \ldots$. In 2026 Hanna conjectured that $a(n)$ is odd exactly when n is a power of 2. We first give the infinite nesting a precise meaning: it defines exactly one power series, and cutting it off after d levels gives the correct coefficients up to degree $d + 1$. This holds for arbitrary weights in place of $1, 2, 3, \ldots$. We then prove the conjecture. Modulo 2 the weight 2 at the second level vanishes, the whole nest collapses, and A satisfies the Catalan equation $A = x + A^2$, whose solution modulo 2 is $x + x^2 + x^4 + x^8 + \cdots$. The results have been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [5] is a large public database of integer sequences, used by mathematicians and scientists to identify sequences and to record what is known about them. Many entries also record *conjectures*: patterns that the contributor observed in the data but could not prove.

This paper is about entry A195194 [1], contributed by Paul D. Hanna in 2011. The entry reads, in part:

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G.f.:  A(x) = INV(x - x*INV(x - 2*x*INV(x - 3*x*INV(x - 4*x*INV(x -
...)))))), where INV(F(x)) = series reversion of F(x).

Conjecture:  a(n) is odd iff n is a power of 2.  - Paul D. Hanna, Apr 07
2026
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— OEIS A195194, Paul D. Hanna, September 2011

Here "G.f." stands for *generating function*: the terms $a(1), a(2), a(3), \ldots$ of the sequence are stored as the coefficients of the power series $A(x) = a(1)\,x + a(2)\,x^2 + a(3)\,x^3 + \cdots$.

Series reversion is the power-series version of an inverse function. If $F(x) = x + f_2x^2 + f_3x^3 + \cdots$, then $\mathrm{INV}(F)$ is the series $V(x) = x + v_2x^2 + \cdots$ for which substituting V into F gives back x: $F(V(x)) = x$. For example, $\mathrm{INV}(x - x^2)$ is the series $x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$ of the Catalan numbers. Section 2 explains this in detail. The entry builds A from an infinite tower of such reversions: in the notation of its examples, $A = \mathrm{INV}(x - x\,F_1)$, where $F_1 = \mathrm{INV}(x - 2x\,F_2)$, where $F_2 = \mathrm{INV}(x - 3x\,F_3)$, and so on without end. It is not obvious what such an infinite tower means; Section 3 gives it a precise meaning.

The first terms, starting at $n = 1$, are 1, 1, 4, 29, 312, 4528, 83736, 1899663, 51453040, ... They grow very quickly.

1.2 The question

Hanna's conjecture says that $a(n)$ is odd exactly when n is one of $1, 2, 4, 8, 16, \ldots$. The table shows the parity of the first terms (1 means odd, 0 means even). The powers of 2 are printed in bold.

n	**1**	**2**	3	**4**	5	6	7	**8**	9	10	11	12	13	14	15	**16**
$a(n)$ mod 2	**1**	**1**	0	**1**	0	0	0	**1**	0	0	0	0	0	0	0	**1**
n	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	**32**
$a(n)$ mod 2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	**1**
n	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
$a(n)$ mod 2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

1.3 What this paper proves

Main results. *(1) The infinite tower of reversions defines exactly one series, and cutting it off after d levels gives the correct coefficients up to degree $d + 1$; this holds for arbitrary weights in place of $1, 2, 3, \ldots$ (Theorem 3.1). (2) The coefficient $a(n)$ is odd if and only if n is a power of 2 (Theorem 3.2).*

The reason for (2) is that, modulo 2, the weight 2 at the second level of the tower vanishes. This cuts the tower off after its first level, and what remains is the equation $A = x + A^2$, which defines the Catalan numbers. Their behaviour modulo 2 is classical.

1.4 What was known

The sequence was added to the OEIS in 2011, and the conjecture in April 2026. The entry gives no proof. Its computer program evaluates a tower of finite height, which Theorem 3.1

shows is correct. The parity of the Catalan numbers is classical; see for example [2]. We are not aware of an earlier proof of the conjecture.

1.5 Why it is interesting

Infinitely nested expressions are a common way of describing power series in the OEIS, and they raise a natural question: what does the expression mean? Our first theorem answers this for every tower of reversions of this shape: each additional level fixes one more coefficient, so the coefficients of the truncated towers eventually stop changing. The parity proof then shows a pleasant phenomenon: working modulo a number p can turn an infinite object into a finite one, because the weight p in the tower becomes 0. Section 6 explains what this gives for other primes p.

1.6 How to read this paper

Section 2 explains the notions we need: power series, substitution and reversion, remainders, and the Catalan numbers modulo 2. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the result and some open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Power series as bookkeeping

A *formal power series* is a way of storing an infinite sequence of numbers $u_0, u_1, u_2, \dots$ in one object,

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots .$$

The letter x is only a placeholder that keeps the terms apart. We never substitute a number for x, so the question of whether the series "converges" never arises; a power series is simply its list of coefficients. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U. Two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum, so everything is well defined. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, because all the other terms cancel. So $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, and we write it as $\frac{1}{1-x}$. In the same way, every series D whose constant term is 1 has exactly one multiplicative inverse $\frac{1}{D}$. Its coefficients can be computed one at a time from $D \cdot \frac{1}{D} = 1$, using only additions, subtractions and multiplications. Every fraction in this paper has a denominator of this kind, so it stands for an honest power series.

Two more pieces of notation. We write $U = O(x^d)$ when all coefficients of U below degree d are zero. So $U - V = O(x^d)$ means that U and V agree up to degree $d-1$. A series of the form $x + u_2 x^2 + u_3 x^3 + \cdots$, with constant term 0 and coefficient of x equal to 1, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1 V(x) + u_2 V(x)^2 + u_3 V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Undoing a substitution

A normalized series can always be "undone" by another normalized series.

Lemma 2.1 (Compositional inverse)**.** *For every normalized series U there is exactly one normalized series V with $U(V(x)) = x$, and this V also satisfies $V(U(x)) = x$. If U has integer coefficients, so does V.*

We call V the *compositional inverse* of U. The OEIS calls it the *series reversion* of U. For example, the compositional inverse of $x - x^2$ is $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, whose coefficients are the Catalan numbers. Indeed $C - C^2 = x$, which says exactly that substituting C into $x - x^2$ gives x.

Proof. Write $V = x + v_2 x^2 + v_3 x^3 + \cdots$ with unknown coefficients. In $U(V) = V + u_2 V^2 + u_3 V^3 + \cdots$, the coefficient of x^n (for $n \ge 2$) equals v_n plus terms that involve only $v_2, \ldots, v_{n-1}$. Setting it equal to 0 therefore determines v_n from the earlier coefficients, using only additions, subtractions and multiplications. So there is exactly one V with $U \circ V = x$, and it has integer coefficients if U does. Applying the same construction to V gives a normalized W with $V \circ W = x$. Then $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, so $V \circ U = V \circ W = x$ as well. □

2.4 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. For example, $17 \equiv 5 \pmod 4$, since both leave remainder 1, and $-3 \equiv 1 \pmod 4$. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$. Saying that a is odd is the same as saying $a \equiv 1 \pmod 2$.

Residues can be added and multiplied consistently, as on a clock. The residues $0, 1, \ldots, m-1$ with this "clock arithmetic" form a number system written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so adding and subtracting are the same thing.

For power series with integer coefficients, $U \equiv V \pmod m$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is simple:

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute with clock arithmetic. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

A word on "rings". Several lemmas below hold not only for integers, but in every number system in which one can add, subtract and multiply with the usual rules (such as $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{Z}/m\mathbb{Z}$). Mathematicians call such a number system a *commutative ring*. We use the word only in this sense. No division is ever needed, which matters because in $\mathbb{Z}/m\mathbb{Z}$ one cannot always divide (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.5 The method in one paragraph

Almost every proof in this paper follows the same three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including the clock arithmetic $\mathbb{Z}/2\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo 2 gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A good guess.* Find a simple series that satisfies the reduced equation and check that it does. By uniqueness, this simple series *is* the reduction of the sequence, and its coefficients can simply be read off.

The creative part is step 3: finding a simple series that works. Step 1 sometimes needs care, when the equation seems to require a division. For a first example, take the Catalan series $C = x + C^2$ from above. Its coefficients $1, 1, 2, 5, 14, 42, \ldots$ are determined one at a time by this equation. Modulo 2, the series $S = x + x^2 + x^4 + x^8 + \cdots$ satisfies the same equation: in $\mathbb{F}_2$, $S^2 = x^2 + x^4 + x^8 + \cdots$ (see Lemma 2.2), so $x + S^2 = S$. Hence $C \equiv S \pmod 2$, and the Catalan coefficient $[x^n]\,C$ is odd exactly when n is a power of 2.

2.6 Squaring in clock arithmetic modulo 2

Lemma 2.2 (Frobenius identity)**.** *Let p be a prime. For every power series U with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

Proof. In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by p. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. Only finitely many terms matter for any given coefficient, so this is a polynomial identity. □

For $p = 2$ this says that squaring a series modulo 2 just doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod 2$. In particular, the square of any series has, modulo 2, no terms of odd degree.

2.7 The Catalan numbers modulo 2

Lemma 2.3. *In every commutative ring the equation $U = x+U^2$ has exactly one solution with $U(0) = 0$. Over the integers it is the Catalan series $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, and*

$$C(x) \equiv x + x^2 + x^4 + x^8 + \cdots = \sum_{k\geq 0} x^{2^k} \pmod 2.$$

So $[x^n]\,C$ is odd exactly when n is a power of 2.

Proof. The coefficient of x^n in U^2 involves only $u_1, \ldots, u_{n-1}$, so the equation determines u_n one at a time: $u_1 = 1$ and $u_n = \sum_{i=1}^{n-1} u_i u_{n-i}$. Over $\mathbb{F}_2$, the series $S = \sum_k x^{2^k}$ satisfies $x + S^2 = x + \sum_k x^{2^{k+1}} = S$ by Lemma 2.2, so by uniqueness $C \equiv S \pmod 2$. □

The parity of the Catalan numbers is classical; see for example [2]. (Here the Catalan number $\frac{1}{n}\binom{2n-2}{n-1}$ appears as the coefficient of x^n, so "n a power of 2" corresponds to the familiar statement that the k-th Catalan number is odd exactly when $k+1$ is a power of 2.)

3 Results

Following the OEIS, we write $\mathrm{INV}(F)$ for the compositional inverse of a normalized series F (Lemma 2.1).

Towers of reversions. Let R be a commutative ring, and let $w_0, w_1, w_2, \ldots$ be elements of R, called the *weights*. A *solution of the tower* is a sequence of normalized series $F_0, F_1, F_2, \ldots$ over R such that

$$F_j = \mathrm{INV}\big(x - w_j\, x\, F_{j+1}\big) \qquad \text{for every } j \geq 0. \tag{1}$$

Then $F_0 = \mathrm{INV}(x - w_0 x F_1) = \mathrm{INV}\big(x - w_0 x\,\mathrm{INV}(x - w_1 x F_2)\big) = \cdots$. For OEIS A195194 the weights are $w_j = j+1$, so F_0 is the series in the definition of the entry. The *tower of height d* is obtained by replacing the innermost $\mathrm{INV}(\cdots)$ by x after d levels: we define $F_j^{[d]}$ for all j by

$$F_j^{[0]} = x, \qquad F_j^{[d+1]} = \mathrm{INV}\big(x - w_j\, x\, F_{j+1}^{[d]}\big).$$

For example, $F_0^{[2]} = \mathrm{INV}\big(x - w_0 x\,\mathrm{INV}(x - w_1 x^2)\big)$.

Theorem 3.1 (The tower is well defined)**.** *For every commutative ring R and all weights $w_0, w_1, \ldots$ in R, the tower (1) has exactly one solution. Moreover, the towers of finite height approach it: $F_j^{[d]} - F_j = O(x^{d+2})$ for all $d, j \geq 0$. That is, $F_j^{[d]}$ and F_j have the same coefficients up to degree $d+1$.*

So the infinite tower in the definition of the entry has exactly one meaning, and the sequence A195194 is $a(n) = [x^n]\,F_0$ for the weights $w_j = j+1$ over the integers.

Theorem 3.2 (Hanna's conjecture)**.** *For the weights $w_j = j+1$ over the integers, the coefficient $a(n) = [x^n]\,F_0$ is odd if and only if n is a power of 2. More precisely,*

$$F_0(x) \equiv x + x^2 + x^4 + x^8 + \cdots = \sum_{k\geq 0} x^{2^k} \pmod 2.$$

4 Proofs

The idea in brief. Each reversion in the tower can be rewritten as an ordinary equation: $F_j = \mathrm{INV}(x - w_j x F_{j+1})$ says that $F_j = x + w_j\, F_j \cdot F_{j+1}(F_j)$. The right-hand side, in degree n, only involves coefficients of degree less than n. So all the series F_j can be built at the same time, one degree at a time; this gives existence and uniqueness. Modulo 2, the weight $w_1 = 2$ is 0, which forces $F_1 = x$; then the equation for F_0 becomes the Catalan equation $F_0 = x + F_0^2$.

Lemma 4.1. *Let U and U' be normalized series that agree below degree r, that is, $U - U' = O(x^r)$. Then their compositional inverses $V = \mathrm{INV}(U)$ and $V' = \mathrm{INV}(U')$ also agree below degree r.*

Proof. Since $U(V) = x = U'(V')$, we have $U(V) - U(V') = U'(V') - U(V') = (U' - U)(V')$. The series $U' - U$ has no terms below degree r, and V' starts with x, so $(U' - U)(V') = O(x^r)$. Now suppose that $V - V'$ were not $O(x^r)$, and let $s < r$ be the lowest degree in which V and V' differ. Then

$$U(V) - U(V') = (V - V') + \sum_{k \geq 2} u_k \left(V^k - V'^k\right).$$

Each $V^k - V'^k = (V - V')(V^{k-1} + V^{k-2}V' + \cdots + V'^{k-1})$ has no terms below degree $s + k - 1 \geq s + 1$. So the right-hand side has a nonzero coefficient in degree s, coming from $V - V'$. This contradicts $U(V) - U(V') = O(x^r)$. □

Proof of Theorem 3.1. Rewriting the tower. Let F_j and F_{j+1} be normalized. The series $x - w_j x F_{j+1}$ is normalized too. By Lemma 2.1, $F_j = \mathrm{INV}(x - w_j x F_{j+1})$ holds if and only if substituting F_j into $x - w_j x F_{j+1}$ gives x, that is, $F_j - w_j\, F_j \cdot F_{j+1}(F_j) = x$, or

$$F_j = x + w_j\, F_j \cdot F_{j+1}(F_j). \tag{2}$$

Existence and uniqueness. Both factors of the product $F_j \cdot F_{j+1}(F_j)$ have no constant term. So the coefficient of x^n in this product involves only coefficients of F_j and F_{j+1} of degree less than n. Hence the whole sequence $F_0, F_1, F_2, \ldots$ can be built at the same time, one degree at a time: every F_j has constant term 0 and coefficient 1 in degree 1; and once all the F_j are known below degree n, equation (2) determines the coefficient of x^n in every F_j. This shows that there is exactly one solution. Only additions and multiplications are used, so this works in every commutative ring.

Towers of finite height. We show $F_j^{[d]} - F_j = O(x^{d+2})$ by induction on d, for all j at once. For $d = 0$, both $F_j^{[0]} = x$ and F_j are normalized, so they agree below degree 2. Suppose $F_{j+1}^{[d]} - F_{j+1} = O(x^{d+2})$. Then the series $x - w_j x F_{j+1}^{[d]}$ and $x - w_j x F_{j+1}$ differ by $w_j x\,(F_{j+1} - F_{j+1}^{[d]}) = O(x^{d+3})$. By Lemma 4.1, their inverses $F_j^{[d+1]}$ and F_j also differ by $O(x^{d+3})$. □

Proof of Theorem 3.2. Let $F_0, F_1, \ldots$ be the solution over the integers with $w_j = j + 1$. Reducing all coefficients modulo 2 gives normalized series over $\mathbb{F}_2$ that satisfy (2) with the weights reduced modulo 2, because reduction is compatible with all the operations involved. By the uniqueness in Theorem 3.1 for the ring $\mathbb{F}_2$, these reductions are *the* solution of the tower over $\mathbb{F}_2$. We now work in $\mathbb{F}_2$.

The weight $w_1 = 2$ is 0 in $\mathbb{F}_2$, so (2) for $j = 1$ says simply $F_1 = x$. Then (2) for $j = 0$, with $w_0 = 1$, says

$$F_0 = x + F_0 \cdot F_1(F_0) = x + F_0 \cdot F_0 = x + F_0^2.$$

This is the Catalan equation, and by Lemma 2.3 its solution over $\mathbb{F}_2$ is $\sum_{k\geq 0} x^{2^k}$. So $a(n)$ is odd exactly when n is a power of 2. □

Notice that, modulo 2, all the deeper levels $F_2, F_3, \ldots$ of the tower play no role at all.

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We evaluated the tower with exact integer arithmetic, once with height 143 and once with height 153. The two results agree below degree 140, as Theorem 3.1 predicts. This reproduces all 20 terms printed in the OEIS entry, and the parity rule of Theorem 3.2 holds for every n with $1 \leq n < 140$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [10]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [8] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [9], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A195194.lean` of the repository, in the namespace `FinalCollection.A195194`. There the solution of the tower is called `tail` and the towers of finite height `approximation`; the proofs are in the module `Round4/NestedReversion.lean`. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
Theorem 3.1 for arbitrary weights in any commutative ring	`arbitrary_weights_unique`, `arbitrary_finite_depth`
For $w_j = j + 1$: every F_j is normalized, satisfies both inverse identities, and the solution is unique	`all_tails_normalized`, `all_tails_source`, `nested_definition_unique`
$A = F_0$ is normalized and is the inverse of $x - xF_1$	`normalized_source`, `first_tail_reversion`
The tower of height d is correct up to degree $d + 1$	`finite_nest_coefficients`
Theorem 3.2: $F_0 \equiv x + F_0^2 \pmod 2$, and the parity rule	`binary_equation`, `exact_parity`

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. Theorem 3.2 determines the parity of every term, so it settles the conjecture of the entry completely. It says nothing about the size of $a(n)$, which grows very fast, or about remainders modulo other numbers.

Other primes. The argument of Theorem 3.2 has a general version. Let p be a prime. Modulo p, the weight $w_{p-1} = p$ vanishes, so $F_{p-1} \equiv x$ and the tower becomes finite:

$$F_0 \equiv \mathrm{INV}\Big(x - x\,\mathrm{INV}\big(x - 2x\,\mathrm{INV}(\cdots \mathrm{INV}(x - (p-1)x^2)\cdots)\big)\Big) \pmod{p}.$$

A power series F over $\mathbb{F}_p$ is called *algebraic* if it satisfies a polynomial equation $Q(x, F(x)) = 0$, where Q is a nonzero polynomial in two variables. The innermost series $\mathrm{INV}(x - (p-1)x^2)$ is algebraic; and if U is algebraic, then so is $x - wxU$ for every nonzero weight w, and so is $\mathrm{INV}(U)$ (if $Q(x, U(x)) = 0$, then substituting $V = \mathrm{INV}(U)$ gives $Q(V(x), x) = 0$). So F_0 is algebraic modulo p. By a theorem of Christol, Kamae, Mendès France and Rauzy [3, 4], the remainders $a(n)$ mod p then form a *p-automatic* sequence: they can be computed by a machine with finitely many states that reads the digits of n in base p. For $p = 2$ the machine is very simple: it checks whether n is a power of 2. We have not worked out the machines for $p \geq 3$, and this seems a natural question. (This paragraph is not part of the Lean formalization.)

The method. The proof illustrates a principle used throughout this collection of papers: a sequence defined by an equation between power series often becomes simple after reduction modulo a small number. Here the reduction even turns an infinite construction into a finite one.

About this paper

How the work was done. This paper is one of 34 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [10]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion section was written during this rewrite; the further observations it contains were checked by computer, and they are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [10], where the Lean sources, the numerical checks, and the earlier drafts are also available.

Acknowledgements

The sequence and the conjecture are due to Paul D. Hanna [1].

References

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[10] D. D. Goyal, *Proofs and corrections for 34 OEIS entries, with Lean 4 formalizations*, repository, 2026. `https://github.com/dvgyl/oeis-a396843-proof`.